



TRAINING EXAMPLE — NOT A REAL IEC 62304 DELIVERABLE

This project is a **training site, not a real medical device** under any regulatory definition. The device described in this document — “SentinelFlow 500” — is entirely **fictional**, invented for this course: nothing here is a genuine IEC 62304 deliverable, a real regulatory submission, or evidence of any real organisation’s compliance with anything. It illustrates the *kind* of document a real SaMD/SiMD project would produce. See the [document register](#) for the full explanation.

Clause: 5.5 — Unit Implementation and Verification · **Register:** [Device Example Register](#)

Document ID: SOP-06 · **Device:** SentinelFlow 500 (fictional) · **Safety class:** C
Status: SOP — template (Procedure not yet written for SentinelFlow 500)

Fictional worked example. See [SOP-01](#) for the disclaimer that applies to this whole document set, and the [Device Example Register](#) for the SentinelFlow 500 device description and what “template” status means here.

1. Purpose

This SOP defines how each software unit in SentinelFlow 500's detailed design is implemented and verified against its design specification, per Clause 5.5.

2. Scope

Applies to implementation and unit verification of SentinelFlow 500's control software (Class C — see [SOP-01 §7.3](#)). Covers Clause 5.5 in full, including the additional acceptance criteria (§5.5.4) that apply only at Class C.

3. Responsibilities

ROLE	RESPONSIBILITY UNDER THIS SOP
Software Development Lead	Defines the coding standard; establishes acceptance criteria before implementation
Developers	Implement each software unit to its detailed design; perform or support unit verification
QA/RA	Confirms unit verification evidence is retained as objective evidence

4. Definitions & Abbreviations

TERM	MEANING
Unit verification	Confirming a software unit meets its detailed design, by review, static analysis, testing, or a combination
Acceptance criteria	The pass/fail conditions a software unit must meet before integration, established before implementation for Class C

5. References

- IEC 62304:2006+AMD1:2015, §5.5
- **SOP-05 — Software Detailed Design:** the specification each unit is implemented and verified against
- **SOP-13 — Software Problem Resolution Process:** any anomaly found during unit verification is managed through this process, not fixed silently

6. What the standard requires

REF	TITLE	APPLIES AT CLASS C	WHAT MUST BE PRODUCED
5.5.1	Implement each software unit	Yes	Implemented software units
5.5.2	Establish software unit verification process	Yes	Strategies, methods and procedures for verifying the software units, with test procedures evaluated for adequacy where verification is by testing
5.5.3	Software unit acceptance criteria	Yes	Acceptance criteria for software units, established before integration, and evidence the units meet them
5.5.4	Additional software unit acceptance criteria	Yes	Additional acceptance criteria where present in the design: event sequence, data and control flow, resource allocation, fault handling, variable initialisation, self-diagnostics, memory management and boundary conditions
5.5.5	Software unit verification	Yes	Documented results of software unit verification

(Quoted directly from `applicability.json` , identical in content to the self-referential set's [06](#), whose Status is itself marked **Partial** — that project does not do isolated unit testing of its JavaScript. This device-example set's version is a template for the same clause, not a completed alternative.)

7. Procedure

Not yet written for SentinelFlow 500 — see the [Device Example Register](#) for what "template" status means. Once complete, this section would give SentinelFlow 500's coding standard, its unit verification strategy (review, static analysis, and/or test), and the acceptance criteria for safety-critical units such as dose-rate validation and occlusion detection.

5.5.1 — Implement each software unit

[Template.]

5.5.2–5.5.3 — Unit verification process and acceptance criteria

[Template.]

5.5.4 — Additional acceptance criteria (Class C)

[Template.]

5.5.5 — Unit verification results

[Template — anomalies found here would be logged per SOP-13.]

8. Records

RECORD	PRODUCED BY	RETAINED IN
Coding standard	Software Development Lead	Project technical file
Unit verification results, per unit	Developers	Project technical file

9. Revision History

Illustrative only — SentinelFlow 500 has no real revision history.

VERSION	STATUS	DESCRIPTION
1.0	Template	Structure and requirements table complete; Procedure section pending

